

## A Self-Closing Gate Conundrum

### #1 Victory Royale

#### Element E: Application of STEM Principles and Practices

Student G's family needs a self-closing mechanism that closes their family's gate on its own because his family does not want unwanted animals and people to get inside their fence gate because it is a safety concern that could cause injury. This happens because student G's gate can not close on its own.

Aluminum is one of the lightest materials in the entire world. It is lighter than iron by almost three times but is very strong, and extremely flexible. It is also corrosion resistant with a very thin but very strong layer of oxide film, a cool fact is that if the aluminum takes any damage on the surface the oxide layer repairs itself. Aluminum has a low density and that means that it is very easy to move around, it is also easy to bend because it is super strong so that is why it is good for manufacturing. This connects to our design because our prototype is made from aluminum and it needs to be weather resistant and has to be able to withstand the weight of the tension spring pulling on the galvanized hanger strap.

The feedback that Mr. F, head of maintenance for our school, told us was that we need to add something to the spring and clamp system to stop it from rubbing the paint off student G's gate such as rubber, cloth, or foam. Another piece of feedback that Mr. F gave us was that we should make sure that when the gate is closed there should be enough tension to close the gate all the way.

The feedback that we have received from Student G's mother was that the design we have made fits what she wanted in a self-closing gate. They are qualified to review our solution because they live where the solution is going to be implemented.

Our experts are Mr. F and Mr. C. Mr. F is the maintenance manager and does maintenance on campus. Mr. C is the agriculture teacher and has many fences around his farm and he works around fences every day. Mr. C said he thought it was a good idea, he said put something in between so the metal does not scratch the paint off the fence.

#### Works Cited

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